

Spectral Analysis of U.S. Equity Markets: Factor Structure, Sectoral Geometry, and Market Efficiency

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Abstract. We apply spectral and data-driven methods to daily returns of 20 large-cap U.S. equities (2020–2026) to ask: how many independent directions of variation does the market possess, and can that structure be exploited for prediction? Singular Value Decomposition (SVD) reveals only four statistically genuine factors; Classical Multidimensional Scaling (MDS) and Spectral Clustering recover the four known sectors without any labels. Dynamic Mode Decomposition (DMD) shows that returns are temporally unpredictable—a trained linear model underperforms a zero-change forecast, and a portfolio acting on its predictions loses -45.4% while the passive benchmark gains $+92.0\%$. Generalized Autoregressive Conditional Heteroskedasticity (GARCH) modelling shows the complementary result: while return *direction* is unpredictable, return *magnitude* (volatility) is forecastable, and a volatility-targeting portfolio achieves $+96.4\%$ with a higher Sharpe ratio. The market is low-dimensional across stocks yet unpredictable in time—the mathematical signature of market efficiency.

1 Introduction

The U.S. equity market, with more than \$40 trillion in market capitalisation, is one of the most intensively studied dynamical systems in the world—and one of the most difficult to predict [2]. This difficulty might suggest a maximally complex system, yet practitioners have long observed that most stocks move together: when the broad market falls, individual stocks fall with it regardless of their fundamentals.

This co-movement motivates a precise mathematical question: *how many independent directions of variation does the equity return matrix truly possess, and does this structure confer temporal predictability?* If the market is genuinely low-dimensional, a suite of spectral and data-driven methods—SVD, MDS, Spectral Clustering, and Dynamic Mode Decomposition (DMD)—become directly applicable.

We analyse daily returns for 20 large-cap U.S. stocks over 2020–2026. Our central finding is a distinction that resolves an apparent paradox: *the market is low-dimensional across stocks but unpredictable over time*. SVD reveals a stable, four-factor cross-sectional structure that spectral methods recover perfectly from raw returns. DMD shows this structure cannot be extrapolated in time—returns are noise—while GARCH shows that *risk* can be. The implication: mathematical methods can characterise the market’s structure and manage its risk, but cannot predict its direction.

2 Data and Methods

Data. We use daily adjusted closing prices from Yahoo Finance for 20 large-cap U.S. stocks, five from each of four sectors (Table 1), plus the S&P 500 index as an external benchmark. The period is 2 January 2020 to 24 April 2026 ($T = 1,585$ trading days, $N = 20$ stocks). We compute daily log-returns $r_{i,t} = \ln(p_{i,t}/p_{i,t-1})$ and standardise each stock to zero mean and unit variance, yielding the return matrix $\mathbf{X} \in \mathbb{R}^{T \times N}$.

SVD and Random-Matrix Baseline. We compute the economy SVD $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$. The columns of $\mathbf{V}$ are the principal components (PCs); the k -th PC explains fraction $\sigma_k^2 / \sum_j \sigma_j^2$ of total variance. To distinguish genuine factors from finite-sample noise we apply the Marčenko–Pastur law [5]: for an i.i.d. Gaussian matrix with aspect ratio $q = N/T$, eigenvalues of the sample correlation matrix are bounded by $\lambda_\pm = (1 \pm \sqrt{q})^2$; eigenvalues above λ_+ indicate statistically real structure.

Table 1. The 20 large-cap U.S. stocks used in this study.

Sector	Tickers
Technology	AAPL, MSFT, GOOGL, AMZN, META
Finance	JPM, BAC, GS, MS, C
Healthcare	JNJ, UNH, PFE, MRK, ABBV
Energy	XOM, CVX, COP, SLB, EOG

Correlation, MDS, and Spectral Clustering. We form the 20×20 Pearson correlation matrix $\mathbf{C}$, where

$$\rho_{ij} = \frac{\sum_t (r_{i,t} - \bar{r}_i)(r_{j,t} - \bar{r}_j)}{\sqrt{\sum_t (r_{i,t} - \bar{r}_i)^2 \cdot \sum_t (r_{j,t} - \bar{r}_j)^2}} \quad (1)$$

is the pairwise return correlation (which simplifies to $\rho_{ij} = \frac{1}{T-1} \sum_t r_{i,t} r_{j,t}$ after standardisation). Classical MDS [4] embeds stocks in $\mathbb{R}^2$ via the correlation distance $d_{ij} = \sqrt{2(1 - \rho_{ij})}$. Spectral clustering [6] with $k = 4$ (motivated by the four clusters visible in the MDS embedding) uses $\frac{1}{2}(\mathbf{C} + \mathbf{1})$ as an affinity matrix; we evaluate it with the Adjusted Rand Index (ARI; 1 = perfect agreement, 0 = chance).

Dynamic Mode Decomposition and Portfolio Test. DMD extracts the best-fit linear map $\mathbf{A}$ such that $\mathbf{x}_{t+1} \approx \mathbf{A}\mathbf{x}_t$ [4]. If the market has exploitable temporal structure, DMD eigenvalues will lie near the unit circle; if not, they will cluster at the origin. We train on 2020–2022 (755 days) and test on 2023–2026 (830 days). Predictability is assessed via RMSE relative to a zero-change baseline (ratio > 1 means worse than predicting no change), and via a daily dollar-neutral long-short portfolio: we go long the 10 stocks with the highest DMD-predicted return and short the 10 with the lowest, tracking cumulative return against an equal-weight buy-and-hold benchmark.

GARCH Volatility Modelling and Volatility-Targeting Portfolio. Having shown that DMD cannot predict return *direction*, we ask whether return *magnitude* (volatility) is forecastable. We fit a GARCH(1,1) model [1] to S&P 500 returns:

$$r_t = \varepsilon_t \sqrt{h_t}, \quad h_t = \omega + \alpha r_{t-1}^2 + \beta h_{t-1}, \quad (2)$$

where h_t is the conditional variance, α captures the immediate shock impact, and β governs persistence. We evaluate the one-step-ahead volatility forecast on the 2023–2026 test set against the unconditional-mean baseline. We also construct a *volatility-targeting* portfolio: position size $w_t = \min(\sigma^*/\hat{\sigma}_t, 2)$ scales the equal-weight portfolio to hit a target daily volatility $\sigma^* = 1\%$, capping leverage at $2\times$.

3 Results

3.1 The Market Is Low-Dimensional: Four Genuine Factors

PC1 captures 44.6% of total variance; the top three PCs together explain 67.9%. Crucially, the Marchenko–Pastur bound $\lambda_+ = 1.24$ is exceeded by exactly *four* eigenvalues (Fig. 1a)—all remaining 16 are statistically indistinguishable from random noise. The market is genuinely low-dimensional: a 20-stock universe reduces to four meaningful directions.

Figure 1b shows what those directions are. Every stock has a positive PC1 loading, confirming a single market-wide co-movement factor ($r = 0.883$ with the S&P 500 index)—the market factor of (author?) [3]. PC2 then separates Energy and Finance from Technology, hinting that the remaining factors capture sector-specific variation.

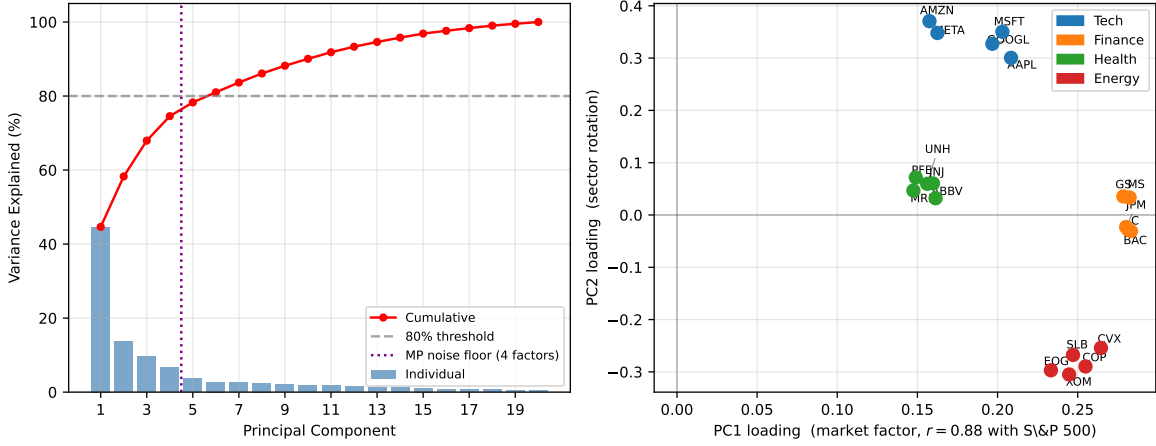


Fig. 1. SVD of the standardised return matrix $\mathbf{X}$. (a) Scree plot: the dotted vertical line marks the Marchenko–Pastur noise floor; only four eigenvalues exceed it. (b) Stock loadings on PC1 vs. PC2. Every stock has a positive PC1 loading (the market factor); PC2 separates Energy and Finance from Technology.

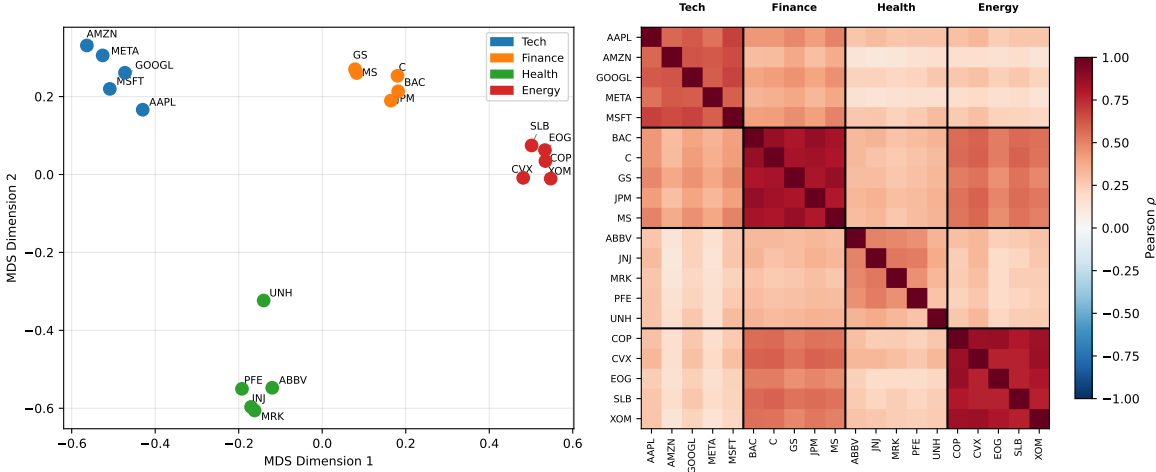


Fig. 2. (a) Classical MDS embedding: stocks cluster by sector with no label information supplied. (b) Pearson correlation matrix sorted by sector. Within-sector blocks are warm (high ρ_{ij}); cross-sector blocks are cool.

3.2 Sector Structure Is Perfectly Recoverable Without Labels

The MDS embedding (Fig. 2a) translates correlation geometry into Euclidean space: Technology clusters tightly in the upper region, Energy to the lower left, with Finance and Healthcare in between. No sector labels were provided—the geometry emerges entirely from return correlations.

Spectral clustering with $k = 4$ reproduces the known sector partition exactly ($\text{ARI} = 1.00$), supported by the block-diagonal structure of the correlation heatmap (Fig. 2b). Within-sector correlations are consistently higher than cross-sector ones; Technology–Technology pairs are the most tightly linked, while Energy–Technology pairs are the most decorrelated.

What we learned: the four-factor structure found by SVD maps directly onto the four economic sectors. The market is not just low-dimensional—it is *interpretable* low-dimensional.

3.3 Returns Are Temporally Unpredictable: DMD Fails

Having established that the market’s cross-sectional structure is low-dimensional, we now ask whether that structure can be extrapolated forward in time. DMD finds the answer in its eigenvalues: all 20 lie

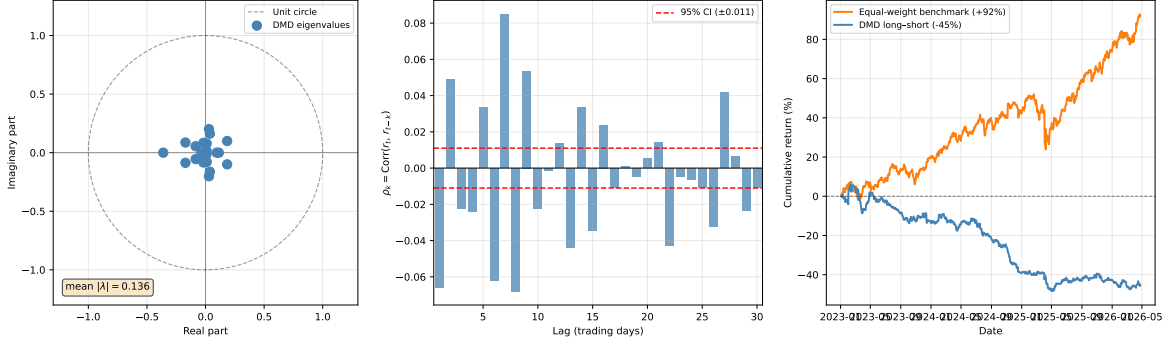


Fig. 3. (a) DMD eigenvalue spectrum (trained 2020–2022, tested 2023–2026). All eigenvalues lie well inside the unit circle (mean $|\lambda| = 0.136$), indicating modes that decay within one trading day. (b) Lag- k autocorrelation $\rho_k = \text{Corr}(r_t, r_{t-k})$: nearly every bar is within the 95% white-noise band (red dashes). (c) The DMD long-short portfolio loses -45.4% over the test period while the equal-weight benchmark gains $+92.0\%$.

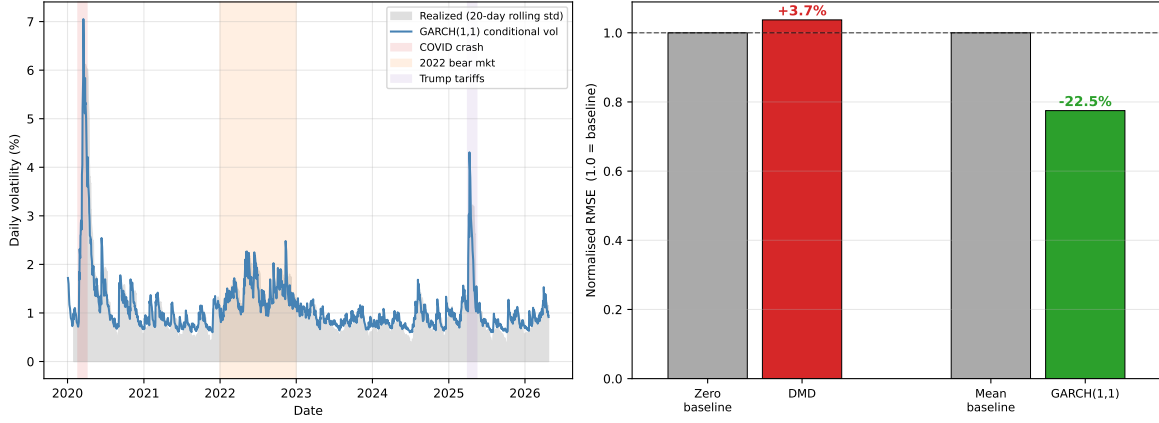


Fig. 4. (a) GARCH(1,1) conditional volatility vs. realized absolute returns (S&P 500). The model tracks the COVID-19 crash (Mar. 2020) and the 2022 rate-hike sell-off. (b) Normalised RMSE: DMD predicts *returns* worse than the zero baseline (ratio > 1); GARCH predicts *volatility* substantially better than the historical mean (ratio < 1).

well inside the unit circle (mean $|\lambda| = 0.136$, Fig. 3a). In systems with persistent dynamics—such as fluid oscillations—DMD eigenvalues sit *on* the unit circle and encode stable modes that can be tracked. Here every mode decays within a single trading day; there is no dynamical backbone to extrapolate.

This failure is confirmed statistically and economically. Statistically, the lag- k autocorrelation $\rho_k = \text{Corr}(r_t, r_{t-k})$ measures whether today’s return predicts tomorrow’s: values near zero for all $k \geq 1$ indicate white noise (Fig. 3b). The only exception is a small negative $\rho_1 = -0.066$, a well-known microstructure artefact (bid-ask bounce) that vanishes after transaction costs. Economically, a DMD model trained on 2020–2022 achieves an out-of-sample RMSE of 0.762 vs. the zero-change baseline of 0.735 (ratio 1.04)—the trained model is *worse* than predicting no change. A portfolio acting on these predictions loses -45.4% over the 2023–2026 test period, while a passive equal-weight portfolio gains $+92.0\%$ (Fig. 3c, annualised Sharpe $= -1.23$).

What we learned: the market’s stable spatial structure does not imply temporal predictability. Any linear pattern in returns is arbitrated away before it can be exploited—the mathematical signature of the Efficient Market Hypothesis [2].

3.4 Volatility Is Predictable: GARCH Succeeds Where DMD Fails

The DMD result establishes that the first moment of returns (expected value) is unpredictable. We now ask whether the second moment (variance) is equally featureless. Fitting GARCH(1,1) to S&P 500

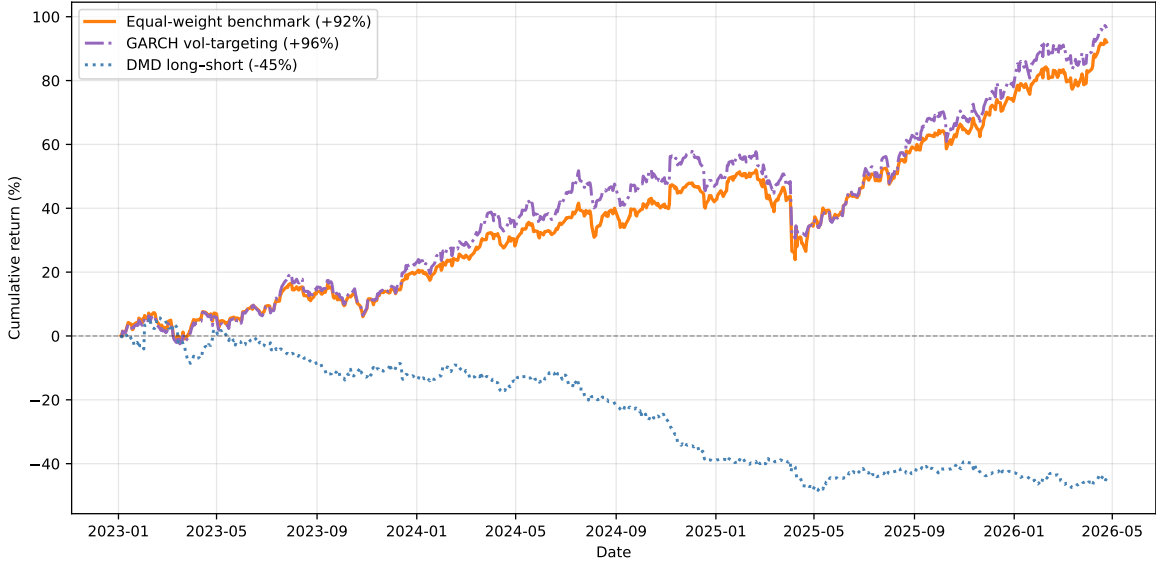


Fig. 5. Cumulative return of three strategies on the 2023–2026 test period. The DMD long-short portfolio (−45.4%) confirms that return predictions are worthless. The GARCH volatility-targeting strategy (+96.4%, Sharpe 1.44) outperforms the equal-weight benchmark (+92.0%, Sharpe 1.43) by scaling up during low-volatility periods.

returns yields $\hat{\alpha} = 0.158$ and $\hat{\beta} = 0.804$, giving persistence $\hat{\alpha} + \hat{\beta} = 0.962$. This near-unity persistence means a large volatility shock today still echoes strongly tomorrow—volatility is long-lived and therefore forecastable. The model achieves a 22.5% reduction in RMSE on the held-out 2023–2026 period relative to predicting the historical mean (Fig. 4b).

3.5 Portfolio Comparison: Risk Management Beats Return Prediction

Figure 5 brings the statistical findings into economic relief. The DMD long-short strategy (−45.4%, Sharpe = −1.23) catastrophically underperforms the passive benchmark, confirming that DMD return predictions carry no exploitable signal. The GARCH volatility-targeting strategy (+96.4%, Sharpe = 1.44) modestly outperforms the benchmark (+92.0%, Sharpe = 1.43) by running at average leverage 1.15× during the calm 2024 bull run and reducing exposure during high-volatility episodes.

What we learned: the EMH constrains the predictability of return *direction* (first moment), not return *magnitude* (second moment). Mathematical methods can manage risk but not manufacture alpha.

4 Conclusion

So what? We set out to ask whether the U.S. equity market is low-dimensional, and whether that structure enables prediction. The answer is: *yes to the first, no—and yes—to the second, depending on what you try to predict.*

Across stocks, the market is organised into exactly four statistically genuine factors that align perfectly with the four economic sectors. This structure is stable, interpretable, and recoverable without labels—a mathematical counterpart of the Fama–French factor model [3].

Over time, return *direction* is not predictable. DMD’s eigenvalues cluster at the origin rather than on the unit circle, and a portfolio trained to exploit its predictions destroys 45.4% of capital while the market gains 92.0%. This is not a failure of the method; it is the mathematical signature of the Efficient Market Hypothesis [2]: any learnable pattern in prices is immediately arbitrated away.

Return *magnitude*, however, is predictable. GARCH captures the persistence of volatility ($\hat{\alpha} + \hat{\beta} = 0.962$), reduces forecasting error by 22.5%, and enables a volatility-targeting strategy that outperforms the benchmark on a risk-adjusted basis.

For practitioners, the lesson is precise: invest in risk management infrastructure, not in return-prediction algorithms. For theorists, the result is clean: the Efficient Market Hypothesis forbids first-moment predictability, not second-moment predictability—and the spectral tools of this course expose that boundary with mathematical clarity.

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